Thms 8 & 10: Proof

Since $a \equiv a' \bmod m$, we have $m \mid (a-a')$, so $a-a' = m \cdot k$ for some $k \in \mathbb{Z}$. Thus $a = a' + mk$. Likewise, $b = b' + m \cdot \ell$ for some $\ell \in \mathbb{Z}$.

We compute $a+b = a' + mk + b' + m\ell$. Thus $a+b - (a'+b') = m(k+\ell)$. Therefore, ~~a+b~~ $a+b \equiv a'+b' \bmod m$. Further, $ab (= a'+mk)(b'+m\ell) \equiv a'b' + mkb' + m\ell a' + m^2 k\ell$.

Thus, $ab - a'b' = m(kb' + \ell a + mk\ell)$ so $ab \equiv a'b' \bmod m$

Suppose:

$a \equiv a' \bmod m$

and $b \equiv b' \bmod m$, then

$a+b \equiv a'+b' \bmod m$ and

$ab = a' \cdot b' \bmod m$

$2x \equiv 7 \bmod 5$

$\cancel{x \equiv \frac{7}{2}}$

$3 \cdot 2x \equiv 3 \cdot 7 \bmod 5$

$6x = 21 \bmod 5$

$x \equiv 1 \bmod 5$

Not every mod has an answer!

mod 8

$\boxed{3^2 \equiv 9 \equiv 1 \bmod 8}$

3 is a soln. to $x^2 - 1 \equiv 0 \bmod 8$

$1^2 \equiv 1 \not\equiv 3 \bmod 4$

$2^2 = 4 \equiv 0 \not\equiv 3 \bmod 4$

$3^2 = 9 \equiv 1 \not\equiv 3 \bmod 4$

$x^2 \equiv 3 \bmod 4$ has no solns

We notice that if $a \equiv b \bmod$, they behave the same as far as m is concerned. Let's just lump all the ~~integers~~ things that behave the same as each other together.

Def 3. Let m be a positive integer. We will define equivalence class relations $\equiv_m$. Define the equivalence class of a mod ~~ab~~ m to be

$$[a]_m = [a] = \bar{a}_m = \bar{a} = a + m\mathbb{Z} = \{x \in \mathbb{Z} : x \equiv a \bmod m\}$$
$$= \{x \in \mathbb{Z} : \mathrm{Rem}(x,m) = \mathrm{Rem}(a,m)\}$$
$$= \{x \in \mathbb{Z} : m \mid (x-a)\}$$

Ex. $[2]_7 = \bar{2}_7$

$= \{x \in \mathbb{Z} : x \equiv 2 \bmod 7\}$

$= \{x \in \mathbb{Z} : 7 \mid (2-x)\}$

$= \{x \in \mathbb{Z} : x = 2 + 7 \cdot k \text{ for some } k \in \mathbb{Z}\}$

$= \{2 + 7k : k \in \mathbb{Z}\}$

$= \{2, 9, 16, 23, 30 \ldots\ldots\}$

$[a]_m = \{a + mk : k \in \mathbb{Z}.\}$

This is a set of integers with ∞ many elements.

$[2]_7 \not\equiv 9 \bmod 7$ $\quad [2]_7 = [9]_7$

Let $a \in [2]_7$ By def $a \equiv \bmod 7$, since $2 \equiv 9 \bmod 7$, we see $a \equiv$

Thus $a \in [9]_7$ Hence, $[2]_7 \subseteq [9]_7$

we denote the set of all equivalence classes mod ab m by $\mathbb{Z}_m$

$\mathbb{Z}_{m\mathbb{Z}} = \{[a]_m : a \in \mathbb{Z}\}$

we call this the set of integers mod ab m. or $\mathbb{Z}$ mod m or $\mathbb{Z}$ mod

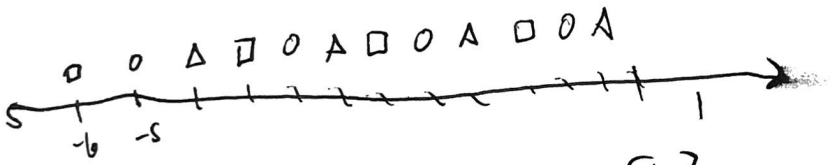
Ex. $\mathbb{Z}/3\mathbb{Z} = \mathbb{Z}_3 = \{[a]_3 : a \in \mathbb{Z}\}$

$= \{[0]_3, [1]_3, [2]_3\}$

$[7]_3 = \{x \in \mathbb{Z} : x \equiv 7 \bmod 3\}$

$= \{x \in \mathbb{Z} : x \equiv 1 \bmod 3\} = [1]_3.$

$7 \equiv 1 \bmod 3$
and $\equiv_3$ is transitive



Thm 12. $a \equiv b \bmod \iff [a]_m = [b]_m$

<u>Proof.</u> If $[a]_m = [b]_m$, note $a \equiv a \bmod m$, so $a \in [a]_m$ thus $a \in [b]_m$, so $a \equiv b \bmod m$. ($\Rightarrow$) Suppose $a \equiv b \bmod m$. Let $x \in [a]_m$, thus $x \equiv a \bmod m$ and $a \equiv b \bmod m$, by the transitivity of congruence mod a b m, we see $x \equiv b \bmod m$. Thus $x \in [b]_m$, and $[a]_m \subseteq [b]_m$. Conversely, let $y \in [b]_m$, thus $y \equiv b \bmod m$ and $b \equiv a \bmod m$, by the transitivity of congruence, we see $y \equiv a \bmod m$. Thus, $y \in [a]_m$, and $[b]_m \subseteq [a]_m$

Cor 13. $[a]_m = [b]_m \iff a \equiv b \bmod m \iff \mathrm{Rem}(a,m) = \mathrm{Rem}(b,m)$

Proof. Thm 12, b & def of $\equiv_m$.

Thm 14. Let $[a] \in \mathbb{Z}/m\mathbb{Z}$ then there is exactly one $b \in \{0, 1 .. m-1\}$ such that $[a] = [b]$.

Proof. Cor 5 + Thm. 12

Cor 15. The rule $x \mapsto [x]_m$ defines a bijection between $\{0, \dots, m-1\}$ and $\mathbb{Z}/m\mathbb{Z}$

Cor 16. $\mathbb{Z}/m\mathbb{Z}$ has m elements. $|\mathbb{Z}/m\mathbb{Z}| = m$

In particular, $\mathbb{Z}/m\mathbb{Z} = \{[0], [1], \dots, [m-1]\}$.